DB-Curator: deep-research agents for scientific database construction

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Abstract

Scientific databases are foundations for reusable knowledge, but their construction remains slow, labour-intensive, and difficult to reproduce across specialised topics. Existing assistants can answer questions or summarise papers, but they rarely produce topic-specific database entries that adapt to open-ended user requests, build the relevant literature set, and preserve entry-level evidence. We present DB-Curator, a multi-agent system that converts a scientific request into a rubric, dynamically constructs a literature set, discovers candidate entities, and verifies each entry through a Researcher–Critique loop. We evaluated DB-Curator in two complementary settings. In release-readiness audits, DB-Curator produced 246/250 release-ready rows and recovered 68 benchmark-missing compounds judged valid by human experts. All 2,166 cited primary-paper references in this audit contained supporting evidence, whereas the zero-shot direct-generation baseline supported only 11/257 references. In benchmark-scale verification over 1,887 supplied candidates, DB-Curator assigned Confirmed labels to 977 entries, compared with 375 for the zero-shot benchmark-labeling baseline. DB-Curator also supported 7,430/7,453 cited references, whereas the strict baseline citation audit supported only 6/2,291 references. Together, these results show that DB-Curator improves both discovery-oriented database release and large-scale evidence-grounded verification. The framework supports scalable scientific curation by coupling dynamic literature construction with predicate-state verification and entry-level provenance.

Main

Scientific databases make experimental findings searchable, comparable, and reusable across studies. Most specialised databases are still built through expert curation. Curators define the database scope, search the literature, extract candidate entries, interpret experimental context, and record evidence for each accepted row. This workflow is powerful, but it is slow, tedious, and difficult to reproduce when the target topic is narrow and evidence is scattered across many papers. In metabolomics and exposomics, resources such as HMDB and the Blood Exposome Database show the value of expert curation. Recent disinfection-by-product (DBP) resources further show how labour intensive topic-specific database construction remains.[1-3] More broadly, resources such as T3DB, PubChem, and DrugBank show that useful databases require entity coverage, stable identifiers, explicit inclusion criteria, and maintainable provenance.[4-6] Scientific database construction is therefore a natural target for automation, provided that automated systems can recover candidates broadly and justify every accepted entry with auditable evidence.

Recent infrastructure makes large-scale scientific curation more technically feasible. Literature search and metadata resources such as Europe PMC, S2ORC, OpenAlex, and related databases expose papers, abstracts, identifiers, and full-text links at scale. Document-processing systems such as PubTator and GROBID support structured parsing of biomedical literature.[7-12] Automated extraction pipelines have also improved substantially. Chemical text-mining tools and benchmarks such as ChemDataExtractor, CHEMDNER, and tmChem support named-entity recognition and normalisation. Domain-adapted models such as BioBERT, SciBERT, and HunFlair2 have strengthened scientific language understanding.[13-18] Together, these resources make it possible to search, parse, and extract from literature collections that are too large for manual inspection alone.

Yet, entity extraction alone is insufficient for database construction. A chemical name mentioned in a paper may be a target compound, reagent, precursor, solvent, contaminant, or contextual comparison. Whether it belongs in a database depends on the user's scope and on evidence about experimental context, sample matrix, analytical confirmation, formation route, organism, or exposure source. These conditions may be distributed across titles, abstracts, methods, results, tables, and supplementary materials. Systems that only extract names can therefore promote plausible but unsupported or off-scope entities into database entries.[19-21] Large language models (LLMs) broaden what is possible in scientific reading and information extraction,[22-24] but direct single-pass use remains risky for database construction. Plausible but unsupported statements can become false entries unless provenance and acceptance rules are enforced explicitly.[25-27]

This makes scientific database construction different from standard retrieval-augmented generation (RAG). A typical RAG system retrieves a context set and generates an answer from it. Database construction instead requires entry-wise decisions. Each candidate must be tested against a user-defined acceptance rule. Missing evidence, exclusion evidence, or off-scope context should change

the entry label, not merely the wording of an answer. The core AI problem is therefore not only how to retrieve relevant papers. It is how to convert a broad scientific request into explicit predicates that can be checked, revisited, and linked to evidence.

Agentic systems provide useful building blocks for this problem. A single-agent workflow can combine search, reading, and decision-making, but these roles are difficult to audit when they are handled by one prompt or one reasoning trace. Multi-agent systems separate these roles. One agent can search and synthesize evidence, while another critiques whether the evidence satisfies the stated rubric. This separation aligns with work on reasoning-and-acting agents, self-correction, deep-research workflows, tool use, and skill composition.[32,33,39-43] Most existing scientific agents, however, focus on question answering, extraction, hypothesis generation, or tool-assisted discovery.[24,34-38] They do not directly address end-to-end scientific database construction, where the literature set, candidate list, evidence criteria, and release decision are all driven by a user-defined database scope.

Here we present DB-Curator, a multi-agent system for on-demand scientific database construction. DB-Curator converts an open-ended user request into a machine-actionable rubric and dynamically constructs the relevant literature set. It then discovers candidate entities, filters them into a verification queue, and evaluates each candidate through a Researcher–Critique loop. Its central design is predicate-state verification. A candidate is accepted only when required rubric predicates are supported by cited evidence and no exclusion criterion is triggered. The output is not a free-form answer or a list of extracted names. It is a provenance-backed database entry that can be inspected, corrected, and reused.

Results

DB-Curator couples dynamic literature construction with predicate-state verification

DB-Curator addresses a central bottleneck in scientific database construction. A useful system must find candidate entities, decide whether each candidate satisfies a user-defined scope, and release each accepted row with traceable evidence. The key design is the coupling of two processes that are usually separated. First, DB-Curator builds a run-specific literature set from the user request, inspected search results, candidate identities, and critique-triggered follow-up searches. Second, it verifies each candidate through an explicit predicate state. Required criteria must be supported, exclusion criteria must not be triggered, and unresolved criteria either direct further evidence search or remain evidence-limited.

Figure 1 summarizes this workflow. DB-Curator first refines the user request into a topic definition and verification rubric. It then performs broad literature harvesting and candidate discovery, filters extracted entities into a smaller verification queue, and applies a Researcher–Critique loop to make entry-wise release decisions. This architecture turns database construction into an evidence-

driven control problem. Each candidate moves from a possible entity to an accepted, excluded, or evidence-limited database row according to the user-defined rubric.

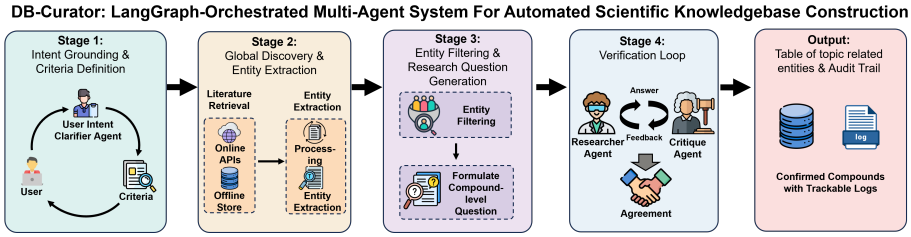


Figure 1 | DB-Curator workflow for on-demand scientific database construction. DB-Curator converts an open-ended user request into a structured, evidence-grounded database through dynamic literature construction and predicate-state verification. The system refines user intent into an acceptance rubric, builds a run-specific literature set, and discovers candidate entities. It then filters candidates into a verification queue and verifies each candidate through a Researcher–Critique loop. Accepted entries are returned with source identifiers, supporting spans, and provenance records.

DB-Curator produces release-ready entries and benchmark-missing discoveries

We first asked whether DB-Curator can produce release-ready database rows from open-ended topic requests. This comparison is needed because a strong zero-shot LLM can often generate plausible compound names. Database construction requires more than plausible names. A released row must have a valid compound identity, satisfy the topic definition, and carry citations that can be traced to supporting primary literature. We therefore compared DB-Curator with a zero-shot direct-generation baseline across five topic-level audits.

A row that matched an existing benchmark entry by normalized identity was counted as a benchmark-overlap row. A non-overlap row was manually inspected against the user-defined topic scope. It was counted as release-ready only when the compound identity and supporting evidence satisfied the topic definition. Citation support was audited at the cited-paper level. A primary-paper reference was counted as supported only when the recovered paper contained the quoted evidence span or equivalent local context.

DB-Curator produced 246/250 release-ready rows across the five fixed cohorts. These included 178 benchmark-overlap rows and 68 non-overlap rows judged valid by human experts. The non-overlap rows are important because they show that DB-Curator is not simply reproducing existing benchmark lists. It can surface literature-supported compounds that are missing from the benchmark but still satisfy the requested database scope. Only four DB-Curator rows were rejected after manual inspection, mainly because of false compound assignments or out-of-scope precursor/product roles.

The evidence audit showed a larger separation between DB-Curator and the zero-shot baseline. All 2,166 cited primary-paper references from DB-Curator resolved to real papers and contained the quoted evidence span or equivalent local context. In contrast, the zero-shot baseline produced many acceptable names under the same overlap-first and manual-inspection rule, but its citation trail was weak. Only 11/257 claimed references could be resolved to real papers with candidate-relevant evidence. Thus, the main advantage of DB-Curator is not name generation alone. It converts generated candidates into auditable database entries with primary-literature support.

The candidate funnel also remained selective. Depending on the topic, DB-Curator evaluated 52–124 compounds to obtain each final 50-row accepted cohort. This indicates that the system does not release every extracted or generated entity. Instead, it filters candidates through topic relevance, compound grounding, and evidence verification before release. Together, these results show that DB-Curator improves discovery-oriented database construction in two ways. It recovers benchmark entries with traceable evidence and identifies valid out-of-benchmark entries that benchmark matching alone would miss.

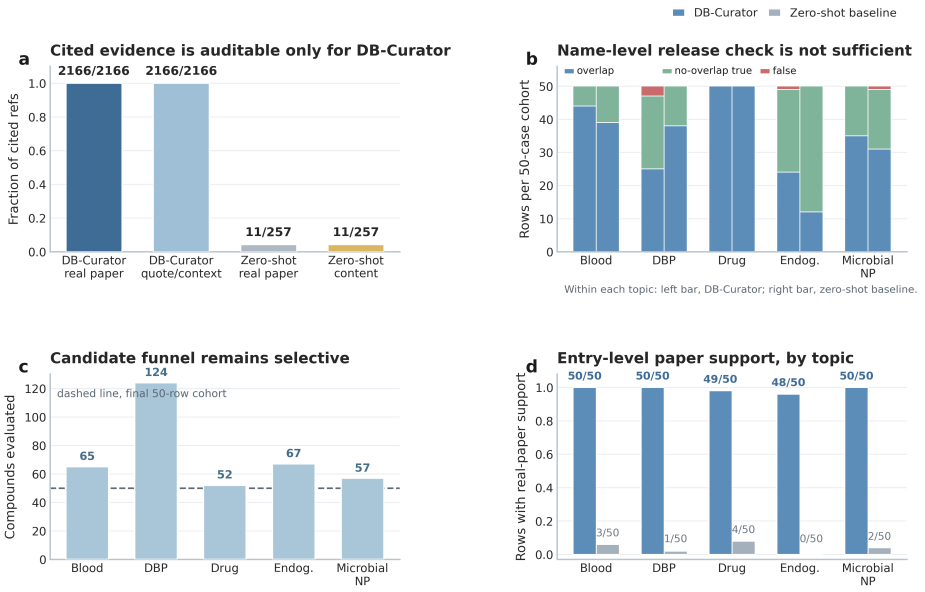


Figure 2 | Evidence grounding, rather than name generation, limits release-ready database construction. A, Reference audit at the cited-paper level, measuring both paper existence and whether the quoted evidence span or equivalent local context was found in the paper. B, Topic-wise name-level release check across the five 50-row cohorts. Within each topic, the left stacked bar is DB-Curator and the right stacked bar is the zero-shot baseline, partitioned into benchmark-overlap, no-overlap true, and false rows. C, Number of per-compound evaluations required to obtain 50 accepted DB-Curator rows in each topic. D, Entry-level

fraction of rows with any real-paper support.

Benchmark-scale verification across five chemical domains

We next tested whether DB-Curator can verify a large supplied candidate set under explicit topic definitions. This setting differs from the release-readiness evaluation. The release-readiness audit asks whether generated rows are suitable for database release, including benchmark-missing discoveries. The benchmark-scale analysis asks whether supplied candidate compounds can be confirmed with direct evidence under the user-defined rubric.

The analysis covered five domains. The DBP set came from a systematic DBP curation resource.[3] The blood exposome, drug, endogenous-metabolite and microbial-natural-product sets came from the Blood Exposome Database,[2] DrugBank,[6] HMDB,[1] and the Natural Products Atlas.[52] The DBP benchmark contained 687 candidate rows. For each of the other four domains, 300 cases were randomly chosen from the corresponding source database. In Figure 3B, only Confirmed rows are counted as database-ready positive entries. Suspected, Explicit Negative Evidence, and Insufficient Direct Evidence are grouped as not confirmed because they do not provide direct positive support for inclusion.

Figure 3A shows the final DB-Curator outcome composition, separating Confirmed rows from entries that remained below the direct positive evidence standard. Figure 3B shows that the final labels were strongly domain-dependent. DB-Curator assigned Confirmed labels to 321/687 DBPs, 86/300 blood exposome chemicals, 210/300 drugs, 123/300 endogenous metabolites, and 237/300 microbial natural products. The zero-shot Gemini 2.5 Flash benchmark-labeling baseline assigned Confirmed labels to fewer cases under the same topic definitions. It assigned Confirmed labels to 137/687 DBPs, 5/300 blood exposome chemicals, 120/300 drugs, 69/300 endogenous metabolites, and 44/300 microbial natural products.

Figure 3C asks whether the cited evidence can support these decisions. DB-Curator cited 7,453 paper references in this benchmark-scale analysis. Of these, 7,433/7,453 resolved to real papers and 7,430/7,453 contained the quoted evidence span or equivalent local context. The matched zero-shot citation-support baseline produced a different failure mode from the direct-generation baseline. In a permissive metadata-resolution audit, many claimed identifiers resolved to some bibliographic record. Title and context checks showed that this was not sufficient evidence support. Under the stricter benchmark-labeling citation audit used in Figure 3C, the zero-shot baseline produced 2,291 claimed references. Only 23/2,291 resolved to title- and identifier-compatible papers, and only 6/2,291 had candidate- and evidence-context support. This shows that the benchmark-labeling task is not just name recognition or citation plausibility. Evidence retrieval and quote-grounded verification are needed to apply topic-specific database criteria.

Figure 3D uses human verification for a different question. It audits rows that DB-Curator resolved as Confirmed or Explicit Negative Evidence and asks whether

the decision aligned with the user-defined topic scope. This matters because a benchmark entry is not automatically inside the narrower user definition used for a particular run. A legacy benchmark may include compounds under broader database conventions, whereas DB-Curator applies the explicit topic definition and rubric supplied for this study. Across 1,091 audited Confirmed or Explicit Negative Evidence rows, 1,085 were judged real, topic-aligned outcomes. Negative labels on benchmark compounds are therefore not necessarily system mistakes. They often indicate that the supplied compound did not satisfy the stricter user topic definition. The full topic definitions and rubric criteria are provided in the Supplementary Information.

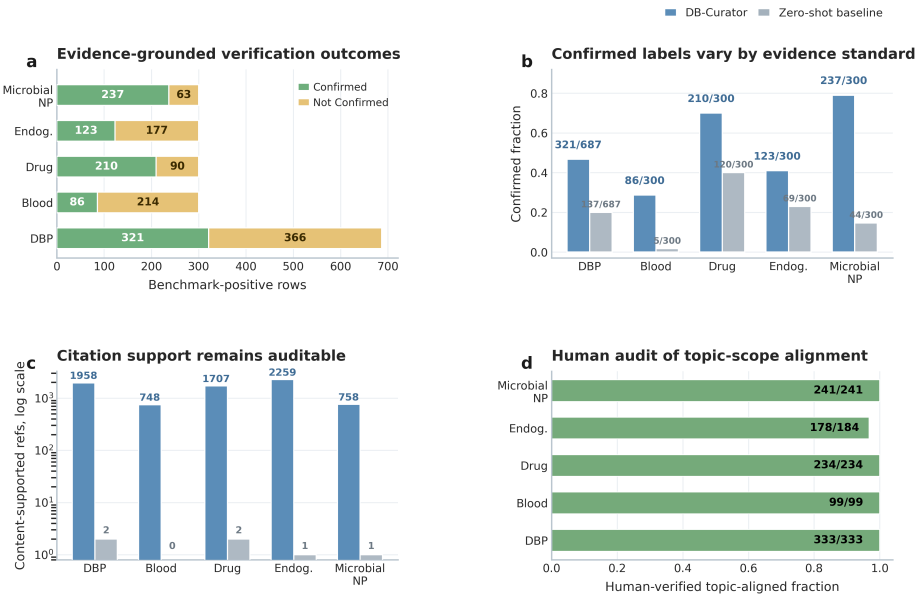


Figure 3 | Benchmark-scale verification, zero-shot baseline, and evidence grounding across five chemical domains. A, DB-Curator benchmark-scale outcome counts, separating Confirmed rows from rows that remained below the direct positive evidence standard. B, Confirmed fractions assigned by DB-Curator and by the zero-shot baseline under the same topic definitions. C, Topic-wise content-supported reference counts for DB-Curator and the matched zero-shot citation-support baseline, shown on a logarithmic scale. D, Human analysis of rows resolved as Confirmed or Explicit Negative Evidence. Counts report how many audited rows were real, topic-aligned outcomes under the user definition.

Rubric discrimination and agent routing

We then asked which rubric gates separate accepted compounds from rows that remain below the direct-evidence threshold. For each topic, we computed overall criterion pass rates and the number of criteria passed by each row. Figure 4 expands the rubric analysis across all five topics rather than collapsing the criteria into one global score. DBP confirmation is driven by direct formation, precursor

identification, and drinking-water relevance. Blood exposome confirmation is driven by evidence for human blood or systemic circulation. Drug confirmation is controlled by drug-entity and formulation-exclusion criteria. Endogenous-metabolite confirmation is driven by human production, function, and exclusion of solely exogenous or microbiota-only explanations. Microbial-natural-product confirmation depends on analytical evidence, unambiguous structural assignment, and microbial origin.

Figure 4 reports how often each rubric gate is passed across evaluated rows for every topic. Green and red markers show the corresponding pass rates in Confirmed and non-Confirmed rows, respectively. Together, these panels make the predicate logic auditable. DB-Curator is not applying a single global confidence score. It routes each topic through the gates that define database readiness for that scientific context. The resulting failure profiles are also informative. Unresolved DBP rows are often missing direct formation evidence, blood exposome rows are often limited by specimen-context evidence, and microbial-natural-product rows depend strongly on structural and origin support.

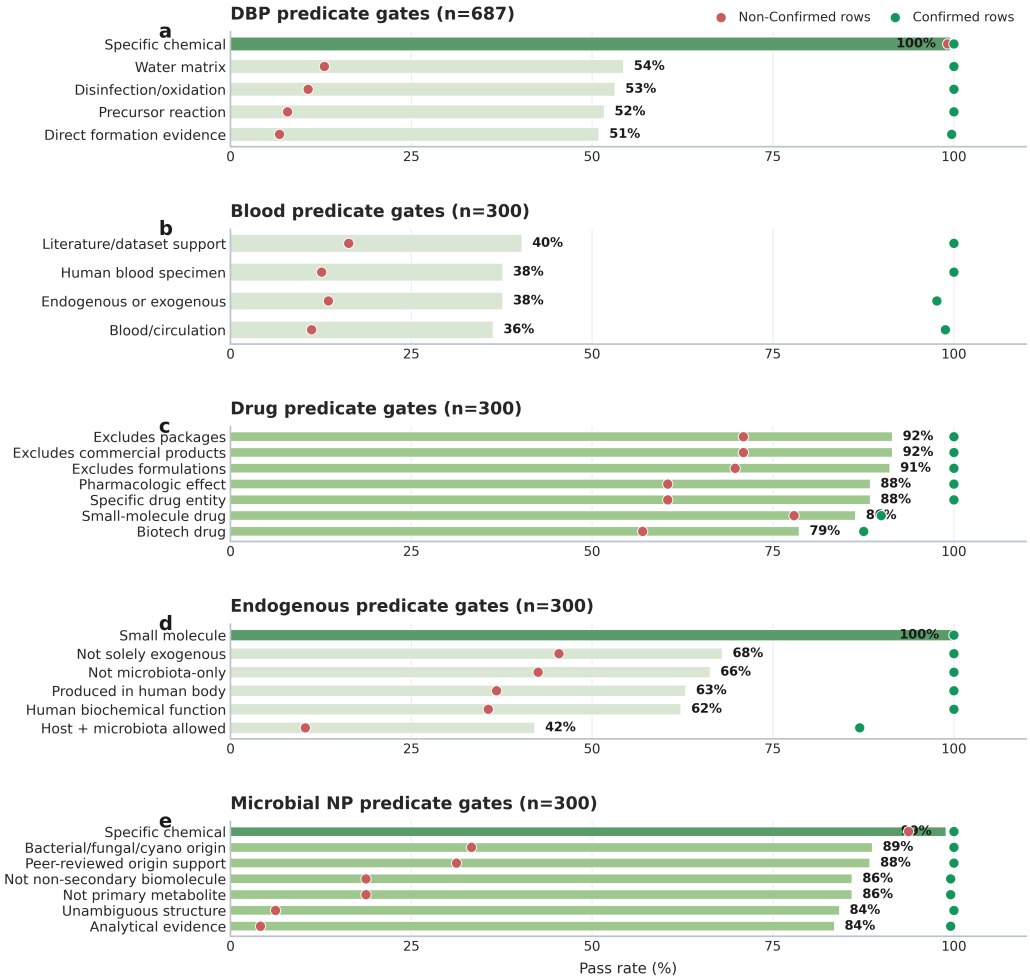


Figure 4 | Topic-specific predicate gates discriminate accepted entries from evidence-limited outcomes. A–E, Rubric diagnostics across the five benchmark topics. Bars report the overall pass rate for each topic-specific rubric gate. Green and red markers indicate pass rates among Confirmed and non-Confirmed rows, respectively. The panels show that database-ready outcomes are separated by topic-specific predicate gates and structured failure patterns rather than a single global score.

Iterative retrieval and Critique-agent utility

Finally, we isolated the role of retrieval mode and iterative critique. The comparisons separate direct generation, topic-definition labeling, researcher-only pre-critique behavior, the full Researcher–Critique loop, and retrieval-source breadth. Direct zero-shot generation produced plausible names but only 11/257 resolvable references. Zero-shot benchmark labeling assigned Confirmed labels to 375/1,887 candidates but yielded only 6/2,291 content-supported references under the strict citation-support audit used for Figure 3C. In contrast, the full Researcher–Critique loop assigned Confirmed labels to 977/1,887 candidates with 7,430/7,453 quote/context-supported references.

Figure 5A treats the first researcher claim as a researcher-only pre-critique state and compares it with the final critique-loop decision. Across 1,887 benchmark rows, the first researcher state assigned Confirmed labels to 933 rows, whereas the final loop assigned Confirmed labels to 977 rows. The loop rescued 49 rows to Confirmed and tightened five initially Confirmed rows away from Confirmed. Figure 5B evaluates these 54 confirmation-changing revisions against human inspection. It shows both topic-wise counts and the pooled 51/54 right-label result. The 49 rescued-to-Confirmed cases were all judged right by human inspection. Among the five tightened-away cases, two became acceptable concrete negative labels. Harmaline, 3-Hexanone, and 6-Succinoaminopurine were over-tightened to Suspected despite human inspection supporting a topic-acceptable final label.

Figure 5C shows that retrieval mode also changes the outcome. In a 50-case mode ablation, hybrid retrieval confirmed 16/50 cases compared with 12/50 for offline-only retrieval. Web and local evidence sources therefore contributed complementary evidence. Hybrid retrieval required more compute, with median runtime increasing from 109 s to 555 s and median token use increasing from 51,024 to 77,343 tokens. Figure 5D shows why additional evidence search can matter. Changed-claim rows required more researcher calls than stable rows, 5.28 versus 2.84 on average. They also required more critique calls, 2.00 versus 1.27, and more supporting sources, 6.40 versus 4.23. These dynamics support the intended role of the Critique agent. As evidence accumulates, the system can revise a provisional claim rather than stopping at the first plausible answer.

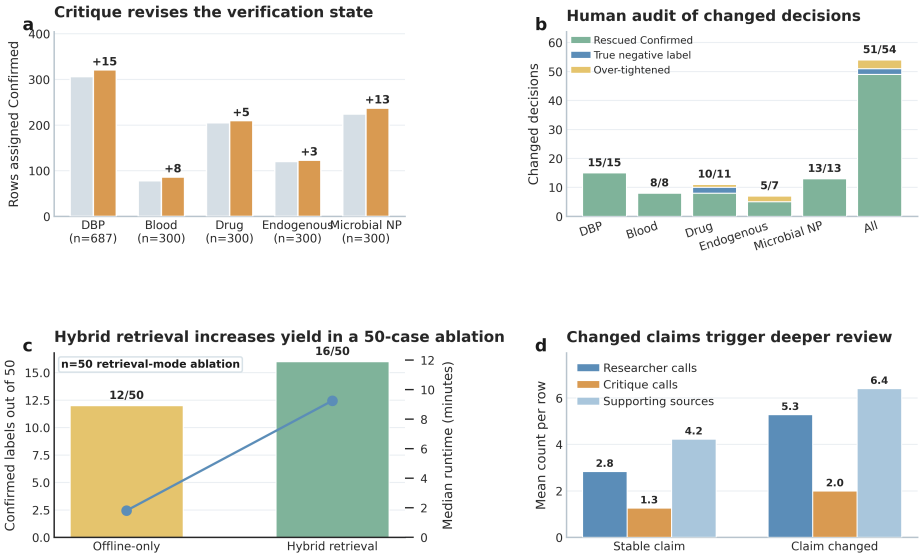


Figure 5 | Iterative retrieval and critique-agent utility. A, Initial researcher claims versus final critique-loop decisions by topic. The left bar is the first researcher claim and the right bar is the final critique-loop decision. B, Human audit of confirmation-changing claim revisions, with topic-wise bars and a pooled All bar separating rescued Confirmed labels, true negative labels, and over-tightened outcomes. C, Offline-only versus hybrid retrieval in a 50-case mode ablation, jointly showing confirmation yield and median runtime. D, Mean exact researcher invocations, critique invocations, and supporting-source depth for rows whose claim stayed stable versus rows whose claim changed during the loop.

Operational cost and routing behavior

We also quantified operational cost because database construction must scale beyond small demonstration sets. The cost profile showed a consistent pattern. Straightforward Confirmed entries usually terminated after shorter evidence paths, whereas evidence-limited and suspected entries consumed more tokens and time. The system continued searching until the relevant rubric gap was resolved or exhausted. Runtime closely followed token use across topics, making token budget a practical proxy for operational cost. Topic-dependent runtime variation reflected differences in literature accessibility and rubric specificity. Rubric score distributions were useful diagnostics, but they did not replace predicate-level adjudication.

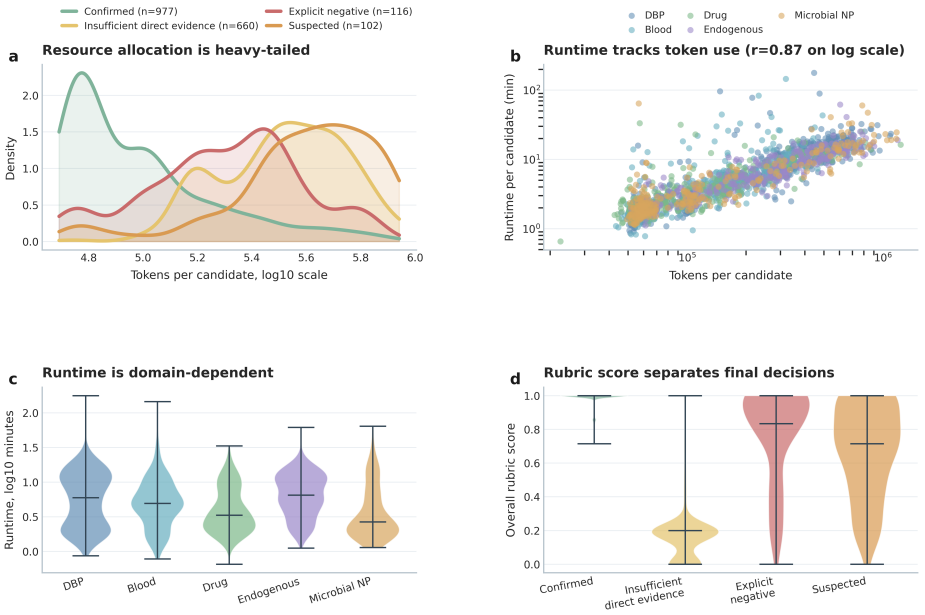


Figure 6 | Operational cost and routing behavior across five chemical domains. A, Smoothed token-use density by final decision, showing heavy-tailed resource allocation. B, Per-row token use versus runtime by topic, showing the close coupling between compute and elapsed time on logarithmic axes. C, Runtime distributions by topic. D, Overall rubric-score distribution by final verdict. In this operational plot, "Insufficient direct evidence" denotes rows that remained below the Confirmed decision threshold.

Discussion

DB-Curator demonstrates that scientific database construction can be formulated as dynamic literature construction plus predicate-state verification. This framing differs from classical chemical information extraction, where the main objective is to detect and normalize entities from a fixed corpus.[13-18,34] It also differs from standard RAG, where retrieved context is usually used to generate an answer. Here, retrieved evidence changes the state of each candidate entry. A candidate can become Confirmed, remain evidence-limited, or be excluded when the retrieved context violates the user-defined rubric.

The closest methodological neighbours are scientific claim verification, agentic reasoning systems, and tool-using scientific assistants. Claim-verification benchmarks emphasize evidence retrieval and rationale selection.[28-31] ReAct, Reflexion, and deep-research agents show that LLMs can interleave reasoning, tool use, and iterative self-correction.[32,33,39,40] MRKL, Toolformer, and Voyager provide broader precedents for modular tool use and skill composition.[41-43] Chemistry-focused systems have shown that LLMs and external tools can support molecular reasoning, extraction, and scientific discovery.[34-38] DB-Curator advances this

line of work by organizing those capabilities around database-entry acceptance. The output is not a free-form answer or extracted mention. It is a structured row whose inclusion criteria, identity grounding, and supporting spans can be audited.

The fixed-cohort results show why this distinction matters. The zero-shot direct-generation baseline produced many acceptable compound names, but its citation trail rarely survived paper-level audit. This is a useful failure mode rather than a surprising weakness. It shows that familiar entity generation and release-quality curation are different tasks. DB-Curator recovered benchmark-overlap compounds, surfaced literature-supported compounds absent from the benchmark, and attached paper-level provenance to accepted rows. The benchmark-scale analyses then show that the same workflow applies across DBPs, blood exposome chemicals, drugs, endogenous metabolites, and microbial natural products. The domain-dependent Confirmed rates are expected because each topic has different evidence requirements and different literature coverage.

The rubric is central to this behaviour. Instead of relying on an implicit model confidence score, DB-Curator converts the user specification into mandatory and exclusionary predicates. The Critique agent then audits claim–evidence alignment at the predicate level. This makes the stopping condition explicit. The loop stops when required evidence is found, when the candidate violates the rubric, or when the available evidence remains insufficient within the search budget. This structure reduces the risk that plausible but unsupported statements are promoted to database facts. It also makes human review easier because reviewers inspect candidate identity, rubric decisions, and cited paper evidence rather than an unstructured generated answer.

The system still has limitations. First, DB-Curator prioritizes text and table evidence. Multimodal extraction from figures and supplementary materials would improve coverage, especially for quantitative measurements. Second, open-access limitations remain important. Some relevant papers cannot be fully inspected unless publisher access, author manuscripts, or user-provided PDFs are available. Future versions could integrate institutionally licensed access, where permitted, or local full-text collections supplied by users. Third, retrieval policy remains a source of performance variance. Improved adaptive query planning, selective follow-up search, tighter coupling between retrieval diagnostics and verification, and stronger penalties for reagents, solvents, and contextual chemicals could improve recall without relaxing acceptance standards. These policies are natural targets for reinforcement learning because downstream verification outcomes provide feedback on which retrieval actions improve evidence quality.

Although the evaluation focuses on chemistry and biomedicine, the architecture is intended to be transferable. Other domains would require new entity recognizers, grounding resources, and rubrics, but the control logic can remain the same. In materials science, predicates could cover composition, synthesis route, electrochemical setting, and reported performance. In clinical-trial curation, predicates could cover intervention, population, endpoint, study design, and statistical

support. From this perspective, DB-Curator is a multi-skill agent workflow for scientific curation. Scope refinement, dynamic literature construction, entity grounding, candidate triage, predicate-level verification, and provenance generation are coordinated around the same acceptance contract. Overall, DB-Curator provides a pathway from on-demand literature retrieval to curated, machine-actionable databases with explicit scope specification, structured evaluation rubrics, multi-agent verification, and auditable provenance.

Methods

Overview

DB-Curator is a stateful multi-agent pipeline that maps a user request u into a curated topic database B. It is designed as a deep research agent rather than as a fixed-corpus extractor. It iteratively searches, reads, and verifies evidence across a paper set that is built during the run. This design can be viewed as agentic retrieval-augmented verification. Retrieval is part of the evolving reasoning state, and new searches are conditioned on the refined user definition, inspected search results, candidate identities, and unresolved rubric predicates. A run maintains a shared state containing the specification, planned queries, retrieved documents, retrieved chunks, candidate entities, evidence snippets, provenance pointers, and audit artifacts. Each node updates this state and writes intermediate outputs to disk so the run can be reproduced and inspected.

$$B = F(u).$$

The pipeline has four stages. Stage 1 performs human-in-the-loop intent refinement and converts the user request into a topic definition, inclusion criteria, exclusions, and verification rubric. Stage 2 performs broad literature harvesting, acquires full-text context when available, and discovers candidate chemicals through named-entity recognition and normalization. Stage 3 converts the noisy candidate set into a smaller verification queue by combining semantic topic similarity, compound grounding, and LLM candidate triage. Stage 4 verifies each candidate through a targeted Researcher–Critique loop that gathers evidence, checks every rubric item, and assigns a canonical claim label.

Definition of specialized agentic tools

DB-Curator executes through a small set of specialized tools with separate roles. These tools implement lower-level skills for literature discovery, chemical grounding, candidate extraction, evidence retrieval, and predicate-level verification. Together, they form the deep-research loop. Broad discovery expands the evidence pool, NER extracts candidate chemicals, and session indexing supports within-paper reading. Iterative querying then changes the paper set and evidence set until a candidate is accepted, excluded, or left evidence-limited.

CompoundDatabaseTool. This tool is invoked only when the verification loop requires compound-specific grounding, for example when a candidate entity is

ambiguous or when additional synonyms are needed to form robust follow-up queries. Its input is a raw chemical string. Its output is a canonical identity and a synonym set from PubChem, with optional properties such as formula and mass when available.

FullTextJournalSearchTool. This tool performs broad literature discovery over online sources and an offline corpus. It searches online sources and offline lexical metadata, then fuses source-specific rankings with reciprocal rank fusion. Its output is a set of retrieved papers with metadata and full-text, JATS, PDF, local-text, or abstract-derived content when available. Because it is called at multiple stages, the retrieved paper set is dynamic: broad topic searches support discovery, candidate-specific searches support entity verification, and critique-generated follow-up queries add papers targeted to unresolved predicates.

ChemicalNerTool. This tool performs chemical named-entity recognition and normalization during the upstream discovery phase. It is invoked after broad retrieval, when many documents have been collected and it would be inefficient to send all raw text directly to an LLM. The tool reads the full-text batch in parallel and distills it into a structured candidate list. It uses HunFlair2 for recognition and a linker module to map surface forms to canonical identifiers such as MeSH and PubChem CIDs. Its input is a directory of retrieved papers or cached text artifacts. Its output is a consolidated report listing identified chemicals, canonical identifiers, frequencies, and source document identifiers. This report supports objective selection and screening of candidates for downstream verification.

FullTextIndexTool. This tool builds a session-specific FAISS index over the full-text artifacts produced in the current run.[46] The output is an index that supports within-document search restricted to the retrieved set.

FullTextQueryTool. This tool performs deep reading for verification over the session index. Its output is a ranked list of evidence snippets with provenance. During verification it is used only for evidence seeking and does not introduce new candidates.

Specification and evaluation criteria refinement

We convert the user request into an operational specification s through a refinement agent. The specification defines inclusion criteria, exclusion criteria, and predicate definitions for what counts as an accepted entry. It is treated as a contract for all downstream agents.

$$s = f_{refine}(u)$$

The specification is then refined into a structured evaluation rubric that defines domain-specific criteria. These criteria can be reused across repeated runs on the same topic so that candidate-level verification is performed against a stable contract. During verification, the Critique agent uses this rubric to make

predicate-level acceptance decisions based on explicitly defined criteria for the domain.

In the current workflow, the human contribution is limited to clarifying scope and resolving ambiguous inclusion rules at the specification stage. Subsequent retrieval, candidate generation, and verification are fully automated under the resulting rubric.

$$r = f_{rubric}(s, \text{domain})$$

For verification, the rubric is treated as a set of predicates rather than as a free-form instruction. For candidate e, each predicate $p_i \in r$ has an evidence state $z_{e,i} \in \{\text{supported}, \text{unresolved}, \text{exclusion-triggered}, \text{exhausted}\}$. Inclusion predicates must be supported, exclusion predicates must not be triggered, and unresolved mandatory predicates drive additional retrieval until the search budget is exhausted. This predicate-state representation is the control signal for the Researcher–Critique loop.

$$z_e = \{z_{e,i} \mid p_i \in r\}$$

A query generation component produces an initial set of broad search directions $Q_0(s)$ from the original query and refined definition. These directions are intentionally recall-oriented and are not converted into a hard domain-specific topic-keyword filter. The upstream agent first searches broad topic phrases, inspects the returned titles and abstracts, mines seed chemical classes and compounds, and then expands searches around those seed terms. This design avoids losing valid papers whose titles or abstracts omit the exact topic wording.

$$Q_0(s) = f_{harvest}(u, s)$$

During candidate verification, queries are generated differently. For a candidate entity e, DB-Curator builds compact structured searches from identity terms I_e, such as the canonical name and a small number of strong synonyms. It combines these with context terms $C_{e,r}$ derived from the current unresolved rubric item. This produces targeted query batches rather than repeated paraphrases of the same broad search.

$$Q_e = f_{verify}(I_e, C_{e,r})$$

Broad retrieval with hybrid online/offline fusion

Broad retrieval is executed by FullTextJournalSearchTool over both online sources and the offline literature corpus. Online sources include PubMed/PMC, Europe PMC, and the Semantic Scholar Academic Graph API,[7,51] with preprint sources enabled only when the run configuration requests them. The offline corpus

contains 6.3 million papers with local metadata and text. Reported full-loop runs used both online and offline retrieval, while the offline-only ablation disabled online APIs. The key distinction from fixed-corpus information extraction is that retrieval is invoked throughout the workflow. The upstream agent first builds a topic-level paper set for candidate discovery, and the downstream verifier later builds candidate-specific paper sets for evidence checking.

For each query q, the tool launches source-specific searches in parallel. Online searches return ranked paper records from their native APIs. Offline search uses SQLite full-text search over title, abstract, identifiers, and metadata.[45] These offline metadata hits are fused with online search results before full-text enrichment.

The retrieval unit at this stage is a paper record rather than a final evidence span. Records are deduplicated by stable paper identity, using DOI, PMID, PMCID, Semantic Scholar identifier, arXiv identifier, or normalized title. Source- and query-specific ranked lists are then combined by reciprocal rank fusion:

$$R(d) = \sum_{j \in J(d)} \frac{1}{k + rank_j(d)},$$

where $J(d)$ is the set of source-query runs that returned document d, $rank_j(d)$ is its rank in run j, and $k = 60$ in the default configuration. The reported full-loop settings kept retrieval broad and passed the fused set to full-text enrichment rather than relying on a narrow per-source cutoff.

$$D_{\text{retrieved}} = Keep_{\text{RRF}}\left(\{d\}, R(d)\right)$$

This dynamic retrieval design makes the literature collection query-conditioned and candidate-conditioned. Early broad searches maximize discovery recall, inspection-derived seed terms expand the candidate-generating corpus, and later critique-derived searches target the specific predicate that remains unsupported for a candidate.

Full text acquisition and parallel processing

For offline full-text acquisition, DB-Curator hydrates retrieved records from local text or PDF paths when those paths are available. For online sources, it first attempts structured JATS or PMC full text, then Unpaywall/open-access PDF routes, and finally PDF conversion. PDF text is extracted with OpenDataLoader[48] as the primary converter and Docling[47] or pdfminer as fallbacks when configured. When full text is unavailable, abstracts are retained as abstract-only context so that the paper can still contribute limited metadata-level evidence. The implementation preserves section headings when possible and supports parallel workers for source search, enrichment, PDF conversion, embedding, and reranking.

Candidate discovery by entity extraction, normalization, and filtering

Candidate discovery is performed after broad retrieval. ChemicalNerTool loads the retrieved paper JSON files in folder mode, extracts text from each file, and processes documents in batches. The NER backend uses HunFlair2 for chemical mention recognition and a biomedical linker for entity normalization. Mentions are associated with source files, and linked identifiers such as MeSH IDs are retained when available. This per-file representation is important because it keeps mention context available for later candidate ranking and audit.

We construct a raw candidate set by aggregating linked entities and surface forms across retrieved papers:

$$E = \bigcup_{d \in D_{\text{retrieved}}} \{norm(m) \mid m \in M(d)\}.$$

For each candidate e, DB-Curator stores mention counts, mention samples, linked identifiers, and surface forms. A representative candidate name is chosen from the observed surface forms and synonyms. When possible, CompoundDatabaseTool checks PubChem compound grounding, uses RDKit for structure handling, and uses OPSIN-based name-to-structure resolution when available.[5,49,50] These checks help prefer names that match concrete compounds over noisy mention strings, identifiers, or broad class labels.

Because the raw candidate pool contains off-topic chemicals, reagents, solvents, precursors, and contextual compounds, DB-Curator applies two filters before expensive verification. The first filter is deterministic pre-ranking. Mention snippets $S(e)$ are scored against the refined topic definition using the BAAI/bge-small-en-v1.5 embedding model, with a TF–IDF fallback when the embedding model is unavailable:

$$s_{\text{topic}}(e) = \max_{s_i \in S(e)} cosine(embed(s_i), embed(s)).$$

The candidate score also includes compound-grounding features from PubChem lookup and mention frequency. Candidates are sorted by topic similarity, grounding score, compound match score, and mention count. The second filter is an LLM candidate-triage step. A Gemini 2.5 Flash candidate-filtering prompt receives only the selected candidate names, synonyms, topic definition, and verification template. It accepts specific compound candidates that are plausibly in scope and assigns a relevance score. Candidates below the score threshold are not sent to downstream verification.

$$E' = \{e \in TopK(E) \mid LLM_{\text{filter}}(e, s) \geq \theta\}.$$

Dual-agent verification and provenance

Each candidate entity $e \in E'$ is verified in its own session. The verification payload contains the candidate name, synonyms, original topic request, refined definition, rubric items, and a compact search budget. The Researcher agent first grounds the entity with CompoundDatabaseTool when possible. This step expands safe synonyms and compound identifiers for search, but it does not add new candidates to the release queue.

Literature discovery during verification uses structured query batches. The query builder prioritizes one or two primary identity terms, then interleaves those identity terms with the current context terms. Additional aliases are kept discoverable but are not crossed with every context term, which prevents synonym explosion. In later rounds, the Critique agent supplies short follow-up queries tied to the highest-priority unresolved rubric item, and duplicate follow-up queries are suppressed.

After papers have been retrieved, FullTextIndexTool builds a session-specific FAISS index. Retrieved papers are split by section when section structure is available, otherwise by full text, using approximately 1200-character chunks with 180-character overlap. Trivial metadata stubs and very short non-substantive chunks are removed before embedding. Chunk embeddings are generated with all-MiniLM-L6-v2.

Evidence seeking is implemented with FullTextQueryTool over this session index. For each generated query, the tool retrieves a larger candidate pool using an ensemble of BM25 lexical retrieval and FAISS vector retrieval with equal weights when both are available. If BM25 is not available, vector retrieval is used alone. The default fetch size is the larger of 120 chunks or four times the requested output size. Candidate chunks are then reranked with BAAI/bge-reranker-v2-m3, with cross-encoder/ms-marco-MiniLM-L-6-v2 as a fallback. Scores are aggregated across query variants by keeping the best score for each chunk, and the top snippets are returned with title, section, DOI, PMID, and source-path metadata.

The Researcher agent synthesizes a claim from these snippets using the provided rubric. The Critique agent then audits whether the claim is supported by the cited evidence and whether every rubric item is solved. The workflow allows several researcher-only rounds before critique if the researcher has not yet produced a fully solved rubric assessment. After critique, one additional targeted researcher round may be executed when the Critique provides actionable follow-up queries. Otherwise, the loop stops when the Critique approves the claim, when the evidence supports rejection, when tools are unavailable, or when repeated follow-up does not add rubric-relevant evidence. To prevent unbounded agent debate, the graph is explicitly bounded. The reported full-loop evaluation runs used a downstream LangGraph recursion limit of 24 graph steps. The Critique prompt also requires one of three actions at each critique point: approve the claim, request a bounded targeted follow-up tied to a specific unresolved predicate, or emit a terminal evidence-limited or negative decision. Duplicate follow-up queries are suppressed, and repeated follow-up without rubric-relevant

evidence forces an exhausted terminal state.

$$y_e = v(e, T_e, r),$$

where T_e is the retrieved and indexed evidence state for entity e, r is the rubric, and y_e is a canonical claim label. The main labels used for analysis are Confirmed, Suspected, Explicit Negative Evidence, and Insufficient Direct Evidence. For accepted entities, DB-Curator stores the candidate identity, final label, rubric assessment, cited papers, and supporting snippets so that the final database can be audited at the evidence-span level.

$$B = \{(e, P_e) \mid e \in E', y_e = \text{Confirmed}\}$$

Evaluation framing

We use two complementary evaluation settings in this manuscript. The first is a fixed-cohort human-audit comparison over 50 DB-Curator confirmed rows and 50 zero-shot baseline rows per topic. In this setting, benchmark overlap is accepted automatically and non-overlap rows are resolved by manual inspection, so the comparison isolates whether each system produces releasable compound entries with auditable citations. The second is a benchmark-scale verification analysis over all available benchmark-positive rows for the five topics. In that setting, the primary outcome is whether each benchmark row reaches Confirmed under the topic-specific evidence rubric, with non-confirmed outcomes collapsed into an Insufficient Direct Evidence class for acceptance-rate analysis. These two settings answer different questions. The fixed cohort measures release readiness of generated entries, whereas the benchmark-scale analysis measures verification behavior over supplied benchmark compounds.

The baselines and ablations are interpreted according to the component they isolate. Because open-ended database construction does not have a standardized retrieve-then-generate protocol, and because retrieval choices would themselves define a new curation system, we use component-isolation analyses rather than an arbitrary retrieve-then-generate baseline. The zero-shot direct-generation baseline tests direct generation from the topic definition without retrieval, compound grounding, or critique. The zero-shot benchmark-labeling baseline tests whether an LLM can apply the same topic definition to supplied benchmark compounds and provide citations. The first researcher claim before critique is used as a researcher-only pre-critique state. The final Researcher–Critique label tests the full predicate-state loop with critique-triggered follow-up retrieval. The offline-only versus hybrid comparison isolates retrieval source breadth while keeping the downstream verification procedure fixed. Together, these comparisons separate direct generation, retrieval breadth, and predicate-level critique rather than treating all improvements as a single retrieval effect.

Citation audit and manual inspection

Citation audits were performed at the cited-paper level. A cited item was first resolved to a primary-paper record using DOI, PMID, PMCID, title, source-path metadata, or publisher, PMC, and PDF routes when available. For each resolved paper, we checked whether the quoted evidence span was present in the recovered text. When exact matching failed because of PDF conversion, formatting, or section-boundary differences, we accepted local-context support only if the same candidate and the same rubric-relevant support meaning were present in the recovered surrounding text. Compound-database identity pages and lookup records were excluded from primary-paper citation counts.

Manual inspections were performed by C.Z. and B.L. using the user-defined topic definitions, rubric criteria, and released audit sheets. Benchmark-overlap rows were accepted automatically, and non-overlap rows were reviewed for topic alignment, false compounds, out-of-scope contexts, and cases where literature supported a row despite a negative LLM comment. These inspections were not blinded to system identity; we therefore release the audit sheets, decision rules, and intermediate overlap and citation files to support external re-review.

LLM settings and reproducibility controls

All LLM calls used fixed prompt templates stored with the experiment code. Unless otherwise stated, generation used deterministic or low-temperature decoding. The evaluated Researcher model, Gemini 2.5 Flash, the Critique model, GPT-5-mini, and the zero-shot benchmark-labeling and citation-support baselines used temperature 0.0. Search budgets, RRF settings, stopping criteria, candidate-filter thresholds, and per-topic definitions were fixed within each reported evaluation run. The full benchmark-scale runs used a downstream recursion limit of 24 graph steps and two consecutive no-progress critique rounds before exhaustion. Per round, the system allowed up to four follow-up query variants and a per-query literature search cap of eight records. It also allowed up to six mechanical full-text search query variants, RRF filtering that retained 30–50 papers, and up to eight passage-query variants for indexed full-text querying. Random subsets for the 300-case benchmark domains and the 50-case retrieval-mode ablation were generated with fixed seeds and released with the evaluation artifacts.

Data availability

The offline literature corpus used in this study is derived from PubMed and Europe PMC and contains 6.3 million full-text articles. The processed offline dataset, including document identifiers and text chunks used for retrieval, is released publicly on Zenodo. Benchmark evaluation results, human audit labels, paper-audit summaries, and overlap analyses are released with the manuscript repository.

Code availability

The source code for DB-Curator is released publicly on GitHub under CZhao97/db-curator. The repository includes the multi-agent orchestration pipeline implemented with LangGraph, hybrid retrieval components,

chemical NER tools with HunFlair2 integration, predicate-level verification logic, and provenance logging. Key files include `code/agent_graph_v2.py` for graph orchestration, `code/prompt_payloads.py` for structured rubric generation, `code/candidate_ranking.py` for semantic relevance filtering, `code/searching_tool.py` for hybrid retrieval over online APIs and the offline corpus, and `manuscript/scripts/` for evaluation and figure-generation workflows. The repository also includes prompts, configuration files for all five evaluation topics, environment specifications, and a browser/backend demo for inspecting toy and precomputed mini-runs.

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Author contributions

C.Z., B.L., X.L., and T.H. conceived the study. C.Z. developed the software system and conducted experiments. C.Z. and B.L. performed the manual result inspection and evaluation analyses. C.Z., B.L., X.L., and T.H. wrote and revised the manuscript. X.L. and T.H. supervised the project.

Competing interests

The authors declare no competing interests.

References

1. Wishart, D. S. et al. HMDB 5.0: the Human Metabolome Database for 2022. Nucleic Acids Res 50, D622–D631 (2022).

2. Barupal, D. K. & Fiehn, O. Generating the Blood Exposome Database Using a Comprehensive Text Mining and Database Fusion Approach. Environ. Health Perspect. 127, 097008 (2019).

3. Xiang, T. et al. Disinfection Byproducts Confirmed over 50 Years: Systematic Curation, Modes of Toxic Action, and Toxicogenomics. Environ. Sci. Technol. 59, 22334–22350 (2025).

4. Lim, E. et al. T3DB: a comprehensively annotated database of common toxins and their targets. Nucleic Acids Res. 38, D781–D786 (2010).

5. Kim, S. et al. PubChem 2023 update. Nucleic Acids Res 51, D1373–D1380 (2023).

6. Wishart, D. S. et al. DrugBank 5.0: a major update to the DrugBank database for 2018. Nucleic Acids Res 46, D1074–D1082 (2018).

7. Levchenko, M. et al. Europe PMC in 2017. Nucleic Acids Res 46, D1254–D1260 (2018).

8. Piwowar, H. et al. The state of OA: a large-scale analysis of the prevalence and impact of Open Access articles. PeerJ 6, e4375 (2018).

9. Lo, K., Wang, L. L., Neumann, M., Kinney, R. & Weld, D. S2ORC: The Semantic Scholar Open Research Corpus. in Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (eds Jurafsky, D., Chai, J., Schluter, N. & Tetreault, J.) 4969–4983 (Association for Computational Linguistics, Online, 2020). doi:10.18653/v1/2020.acl-main.447.

10. Priem, J., Piwowar, H. & Orr, R. OpenAlex: A fully-open index of scholarly works, authors, venues, institutions, and concepts. Preprint at https://doi.org/10.48550/arXiv.2205.01833 (2022).

11. Wei, C.-H. et al. PubTator 3.0: an AI-powered literature resource for unlocking biomedical knowledge. Nucleic Acids Res 52, W540–W546 (2024).

12. Lopez, P. GROBID: Combining Automatic Bibliographic Data Recognition and Term Extraction for Scholarship Publications. in Research and Advanced Technology for Digital Libraries (eds Agosti, M., Borbinha, J., Kapidakis, S., Papatheodorou, C. & Tsakonas, G.) 473–474 (Springer, Berlin, Heidelberg, 2009). doi:10.1007/978-3-642-04346-8_62.

13. Mavračić, J., Court, C. J., Isazawa, T., Elliott, S. R. & Cole, J. M. ChemDataExtractor 2.0: Autopopulated Ontologies for Materials Science. J. Chem. Inf. Model. 61, 4280–4289 (2021).

14. Krallinger, M. et al. The CHEMDNER corpus of chemicals and drugs and its annotation principles. J Cheminform 7 (Suppl 1), S2 (2015).

15. Leaman, R., Wei, C.-H. & Lu, Z. tmChem: a high performance approach for chemical named entity recognition and normalization. J Cheminform 7, S3 (2015).

16. Lee, J. et al. BioBERT: a pre-trained biomedical language representation model for biomedical text mining. Bioinformatics 36, 1234–1240 (2020).

17. Beltagy, I., Lo, K. & Cohan, A. SciBERT: A Pretrained Language Model for Scientific Text. in Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP) (eds Inui, K., Jiang, J., Ng, V. & Wan, X.) 3615–3620 (Association for Computational Linguistics, Hong Kong, China, 2019). doi:10.18653/v1/D19-1371.

18. Sänger, M. et al. HunFlair2 in a cross-corpus evaluation of biomedical named entity recognition and normalization tools. Bioinformatics 40, btae564 (2024).

19. Peng, Y., Rios, A., Kavuluru, R. & Lu, Z. Extracting chemical–protein relations with ensembles of SVM and deep learning models. Database (Oxford) 2018, bay073 (2018).

20. Kononova, O. et al. Opportunities and challenges of text mining in materials research. iScience 24, 102155 (2021).

21. Jain, S., van Zuylen, M., Hajishirzi, H. & Beltagy, I. SciREX: A Challenge Dataset for Document-Level Information Extraction. in Proceedings of the

58th Annual Meeting of the Association for Computational Linguistics (eds Jurafsky, D., Chai, J., Schluter, N. & Tetreault, J.) 7506–7516 (Association for Computational Linguistics, Online, 2020).

22. Dagdelen, J. et al. Structured information extraction from scientific text with large language models. Nat Commun 15, 1418 (2024).

23. Polak, M. P. & Morgan, D. Extracting accurate materials data from research papers with conversational language models and prompt engineering. Nat Commun 15, 1569 (2024).

24. Gupta, S., Mahmood, A., Shetty, P., Adeboye, A. & Ramprasad, R. Data extraction from polymer literature using large language models. Commun. Mater. 5, 269 (2024).

25. Menick, J. et al. Teaching language models to support answers with verified quotes. Preprint at https://doi.org/10.48550/arXiv.2203.11147 (2022).

26. Bohnet, B. et al. Attributed Question Answering: Evaluation and Modeling for Attributed Large Language Models. Preprint at https://doi.org/10.48550/arXiv.2212.08037 (2023).

27. Wu, K. et al. An automated framework for assessing how well LLMs cite relevant medical references. Nat Commun 16, 3615 (2025).

28. Wadden, D. et al. Fact or Fiction: Verifying Scientific Claims. in Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP) (eds Webber, B., Cohn, T., He, Y. & Liu, Y.) 7534–7550 (Association for Computational Linguistics, Online, 2020). doi:10.18653/v1/2020.emnlp-main.609.

29. Wadden, D. et al. SciFact-Open: Towards open-domain scientific claim verification. in Findings of the Association for Computational Linguistics: EMNLP 2022 (eds Goldberg, Y., Kozareva, Z. & Zhang, Y.) 4719–4734 (Association for Computational Linguistics, Abu Dhabi, United Arab Emirates, 2022). doi:10.18653/v1/2022.findings-emnlp.347.

30. Wang, Y. et al. Factcheck-Bench: Fine-Grained Evaluation Benchmark for Automatic Fact-checkers. in Findings of the Association for Computational Linguistics: EMNLP 2024 (eds Al-Onaizan, Y., Bansal, M. & Chen, Y.-N.) 14199–14230 (Association for Computational Linguistics, Miami, Florida, USA, 2024). doi:10.18653/v1/2024.findings-emnlp.830.

31. Zhang, B. et al. A dataset for evaluating clinical research claims in large language models. Sci Data 12, 86 (2025).

32. Yao, S. et al. ReAct: Synergizing Reasoning and Acting in Language Models. in The Eleventh International Conference on Learning Representations (ICLR) (2023).

33. Shinn, N., Cassano, F., Gopinath, A., Narasimhan, K. R. & Yao, S. Reflexion: Language Agents with Verbal Reinforcement Learning. in Advances in Neural

Information Processing Systems 36 (NeurIPS 2023) (2023).

34. Fan, V. et al. OpenChemIE: An Information Extraction Toolkit for Chemistry Literature. J. Chem. Inf. Model. 64, 5521–5534 (2024).

35. Ghafarollahi, A. & Buehler, M. J. ProtAgents: protein discovery via large language model multi-agent collaborations combining physics and machine learning. Digital Discovery 3, 1389–1409 (2024).

36. Ghafarollahi, A. & Buehler, M. J. SciAgents: Automating Scientific Discovery Through Bioinspired Multi-Agent Intelligent Graph Reasoning. Adv. Mater. 37, 2413523 (2025).

37. Chang, C.-Y. et al. MAIN-RAG: Multi-Agent Filtering Retrieval-Augmented Generation. in Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers) (eds Che, W., Nabende, J., Shutova, E. & Pilehvar, M. T.) 2607–2622 (Association for Computational Linguistics, Vienna, Austria, 2025). doi:10.18653/v1/2025.acl-long.131.

38. Bran, A. M., Cox, S., Schilter, O., Baldassari, C., White, A. D. & Schwaller, P. Augmenting large language models with chemistry tools. Nat. Mach. Intell. 6, 525–535 (2024).

39. Huang, Y. et al. Deep Research Agents: A Systematic Examination And Roadmap. Preprint at https://doi.org/10.48550/arXiv.2506.18096 (2025).

40. Xi, Y. et al. A Survey of LLM-based Deep Search Agents: Paradigm, Optimization, Evaluation, and Challenges. Preprint at https://doi.org/10.48550/arXiv.2508.05668 (2025).

41. Karpas, E. et al. MRKL Systems: A modular, neuro-symbolic architecture that combines large language models, external knowledge sources and discrete reasoning. Preprint at https://doi.org/10.48550/arXiv.2205.00445 (2022).

42. Schick, T. et al. Toolformer: Language Models Can Teach Themselves to Use Tools. in Advances in Neural Information Processing Systems 36 (NeurIPS 2023) (2023).

43. Wang, G. et al. Voyager: An Open-Ended Embodied Agent with Large Language Models. Preprint at https://doi.org/10.48550/arXiv.2305.16291 (2023).

44. Mavroudis, V. LangChain v0.3. Preprint at https://doi.org/10.20944/preprints202411.0566.v1 (2024).

45. Hipp, D. R. SQLite. Available at https://www.sqlite.org/ (accessed 3 May 2026).

46. Douze, M. et al. The Faiss library. Preprint at https://doi.org/10.48550/arXiv.2401.08281 (2024).

47. Auer, C. et al. Docling Technical Report. Preprint at https://doi.org/10.48550/arXiv.2408.09869 (2024).

48. OpenDataLoader Project. OpenDataLoader PDF documentation. Available at https://opendataloader.org/docs (accessed 3 May 2026).

49. Landrum, G. RDKit: Open-source cheminformatics. Available at https://www.rdkit.org/ (accessed 3 May 2026).

50. Lowe, D. M., Corbett, P. T., Murray-Rust, P. & Glen, R. C. Chemical Name to Structure: OPSIN, an Open Source Solution. J. Chem. Inf. Model. 51, 739–753 (2011).

51. Kinney, R. et al. The Semantic Scholar Open Data Platform. Preprint at https://doi.org/10.48550/arXiv.2301.10140 (2023).

52. van Santen, J. A. et al. The Natural Products Atlas: An Open Access Knowledge Base for Microbial Natural Products Discovery. ACS Cent. Sci. 5, 1824–1833 (2019).